

MAINS MASTER PROGRAM (MMP) 2024

POLITY – 10

E-GOVERNANCE

Important aspects of Governance, Transparency & Accountability, E-Governance – applications, models, successes, limitations & potential; Citizens Charters; Transparency & Accountability & institutional & other measures

E-Governance – applications, models, successes, limitations & potential

➤ E-governance

- ✓ **E-Governance** refers to the use of information and communication technology (ICT) by government agencies to deliver to enhance the access to and delivery of government services, improve internal operations, and foster greater interaction between the government and its citizens.
- ✓ E-governance aims to make government processes more efficient, transparent, and responsive to the needs of the public.

➤ Stages of Development

- ✓ Computerization
- ✓ Networking
- ✓ On-Line presence
- ✓ On-Line Interactivity

➤ Applications of e-governance

✓ Government to Citizen (G2C)

- **Service Delivery** - Providing public services such as tax filing, bill payments, and obtaining licenses online.
- **Information Dissemination** - Offering easy access to government information, policies, and programs.
- **Citizen Engagement** - Facilitating platforms for citizens to provide feedback and participate in decision-making processes.
- Example – **Voluntary disclosure through RTI, PIB press release, Mygov.in, ITR portal, online FIR etc.**

✓ Government to Business (G2B)

- **Regulatory Compliance:** Simplifying compliance with government regulations through online portals.
- **Procurement and Tendering:** Conducting e-procurement processes to increase transparency and efficiency in government contracting.
- **Support Services:** Offering information and services that help businesses operate more effectively, such as business registration and tax information.
- Example - **e-Procurement; e-bidding; MCA-21etc**

✓ Government to Government (G2G)

- **Interagency Collaboration:** Enhancing communication and collaboration among different government departments and agencies.
- **Data Sharing:** Facilitating the sharing of data and resources to improve policy making and service delivery.
- **Efficiency:** Streamlining internal processes and reducing administrative burdens through the use of technology.
- Example - **PRAGATI; CPGRAMS, Crime & Criminal Tracking Networking System (CCTNS) etc**
- ✓ **Government to Employees (G2E)**
 - **Employee Services:** Providing online access to HR services, payroll, training, and development resources
 - **Communication:** Enhancing communication between government employees and departments.
 - **Performance Management:** Implementing systems to monitor and improve employee performance.
 - Example – **PMS tools, JEEVAN PRAMAAN online initiative etc.**
- ✓ **Hybrid Model**
 - Hybrid models combine elements of the above models to create comprehensive e-governance frameworks that address multiple aspects of governance.
 - Integrated platforms that offer G2C, G2B, G2G, and G2E services.
 - Example - **Digital India Initiative:** An overarching initiative that encompasses various e-governance models to provide a wide range of digital services to citizens, businesses, and government agencies.
- **Digital / Fourth Industrial Revolution**
 - ✓ The Digital Revolution, often referred to as the Fourth Industrial Revolution, signifies a period of rapid technological advancement that is transforming various aspects of human life, including the economy, society, and governance.
 - ✓ It builds on the Third Industrial Revolution, characterized by the rise of computers and digital technology, and introduces a fusion of technologies that blur the lines between the physical and digital spheres.
- **Key Characteristics of the Fourth Industrial Revolution**
 - ✓ **Advanced Technologies**
 - **Artificial Intelligence (AI):** Machines and systems capable of performing tasks that typically require human intelligence.
 - **Big Data and Analytics:** Processing and analysing vast amounts of data to derive actionable insights.
 - **Blockchain:** Decentralized and secure technology for recording transactions and data.

- **Quantum Computing:** Advanced computing technology that leverages quantum mechanics for faster processing power.
- **Robotics and Automation:** Increased use of robots and automated systems in various industries.
- ✓ **Technological convergence**
 - Integration of physical, digital, and biological technologies, creating new possibilities and innovations.
- ✓ **Wide impact on major industries like Manufacturing, Healthcare, Finance & Transportation.**
- ✓ **Societal & Economic changes in areas of Job requirement, Governance, Educational services etc.**
- **Impact of Fourth Industrial Revolution on Public Service Delivery**
 - ✓ **Data Analytics in Public Service**
 - 1) **Data Analytics for Tax Compliance** - The **Income Tax Department** uses data analytics to analyse financial transactions and detect tax evasion. The Project Insight initiative uses data mining and analytics to identify discrepancies in tax filings. This helps in increasing tax compliance, detects, and prevents tax evasion, and enhances revenue collection.
 - 2) **Data Analytics for Smart Cities** - The **Smart Cities Mission** employs data analytics to manage urban infrastructure, optimize traffic flow, and improve public services. Cities like Pune and Bhubaneswar use data-driven insights for better urban planning. This helps in enhancing quality of urban life, improves infrastructure management, and ensures efficient use of resources.
 - 3) **Data Analytics for Targeted Delivery** - The **Pradhan Mantri Jan Dhan Yojana (PMJDY)** uses data analytics to identify eligible beneficiaries and ensure targeted delivery of financial services. Data analytics helps in monitoring the impact of social welfare programs.
 - ✓ **Artificial Intelligence in Public Service**
 - 1) **AI for Crop Management** - The Ministry of Agriculture and Farmers Welfare uses AI for the **Pradhan Mantri Fasal Bima Yojana (PMFBY)** to assess crop damage and estimate insurance claims. AI-based solutions analyse satellite imagery and weather data to provide accurate assessments.
 - 2) **AI for Crime Prevention** - The **Delhi Police** use **AI-based predictive policing** tools to analyse crime data and predict potential crime hotspots. This helps in deploying resources more effectively and preventing crimes before they occur.
 - ✓ **Blockchain technology in Public Service**
 - 1) **Blockchain for Food Distribution** - The Karnataka government is exploring the use of blockchain to streamline the **Public Distribution**

System (PDS) to ensure transparency and efficiency in the distribution of food grains. Reduces leakages, ensures rightful delivery of subsidies, and increases accountability in the distribution process.

2) **Blockchain for Land records management** - The **Andhra Pradesh** government has implemented blockchain technology for land title registration to create a tamper-proof and transparent system. This ensures secure and immutable records of land ownership.

➤ **Challenges & Issues**

- ✓ Digital Divide & Illiteracy
- ✓ Cybersecurity issues
- ✓ Legal and Regulatory Frameworks
- ✓ Data Quality and Integrity
- ✓ Data Privacy and Security
- ✓ Developing Infrastructure

➤ **Impact on Socio-Economic Development due to digital illiteracy & Lack of access to ICT in rural areas.**

- ✓ Digital illiteracy and lack of ICT accessibility in rural areas significantly hinder socio-economic development. This issue is critical as it affects various aspects of life, including **education, healthcare, economic opportunities, and overall quality of life.**
- ✓ **Impact on Education**
 - **Limited Access to Information:** Students in rural areas without ICT access miss out on the vast resources available online, hindering their learning and educational development.
 - **Quality of Education:** Digital tools can enhance teaching methods and provide interactive learning experiences. Without these, education quality remains substandard
 - **Teacher Training:** ICT can offer continuous professional development for teachers. Lack of access means rural teachers may not receive the same quality of training as their urban counterparts.
 - **For example,** during the **COVID-19 pandemic**, many rural students were unable to attend online classes due to lack of devices and internet access, widening the education gap between urban and rural areas.
- ✓ **Impact on Health care**
 - **Telemedicine:** ICT can bring healthcare to remote areas through telemedicine. Lack of access means rural populations miss out on timely medical consultations and treatments.
 - **Health Information:** Digital literacy enables people to access health information, preventative care guidelines, and emergency services online. Without it, health outcomes can be poorer.

✓ **Impact on Economic opportunities**

- **Employment:** ICT skills are increasingly necessary for many jobs. Digital illiteracy limits employment opportunities and restricts rural populations to low-paying, unskilled work.
- **Entrepreneurship:** Access to ICT can help rural entrepreneurs reach broader markets and access online platforms for business growth. Lack of access restricts their market reach and growth potential.
- **Agricultural Productivity:** ICT tools can provide farmers with vital information on weather, prices, and best practices. Without this, agricultural productivity and income remain low.

✓ **Impact on social inclusion**

- **Civic Participation:** Digital literacy and ICT access enable citizens to engage in governance processes, access government services, and participate in democratic activities. Without it, rural populations have less influence and voice in governance.

➤ **Digital India Program**

- ✓ The Digital India Programme is a **flagship initiative** launched by the Government of India in 2015 to transform India into a **digitally empowered society and knowledge economy**.
- ✓ The initiative aims to ensure that government services are made available to citizens electronically by improving online infrastructure and increasing internet connectivity.

✓ **Vision of the Program**

1) Digital Infrastructure as a Utility to Every Citizen

- ❖ Ensuring high-speed internet access for all citizens.
- ❖ Providing a unique digital identity to every citizen.
- ❖ Ensuring mobile and banking services are available to all.

2) Governance and Services on Demand

- ❖ Seamless integration across departments or jurisdictions.
- ❖ Real-time availability of services to citizens.
- ❖ Digitally transformed services to simplify government processes.

3) Digital Empowerment of Citizens

- ❖ Universal digital literacy.
- ❖ Availability of digital resources/services in Indian languages.
- ❖ Collaborative digital platforms for participative governance.

✓ **Nine pillars of Digital India Program**

1. Broadband Highways

- ❖ Establishing nationwide high-speed internet networks.

- ❖ Projects like **Bharat Net** aim to provide broadband connectivity to all Gram Panchayats.

2. Universal Access to Mobile Connectivity

- ❖ Ensuring mobile connectivity to all regions, particularly rural and remote areas.

3. Public Internet Access Programme

- ❖ Transforming post offices into multi-service centres.
- ❖ Expanding Common Service Centres (CSCs) to provide a range of digital services.

4. e-Governance

- ❖ Simplifying and enhancing government processes through IT.
- ❖ Digitization of government records and procedures.

5. e-Kranti - Electronic Delivery of Services

- ❖ Delivering all government services electronically.
- ❖ Initiatives like e-education, e-health, technology for farmers, financial inclusion, etc.

6. Information for All

- ❖ Open data platform to share information and promote transparency.
- ❖ Online hosting of government data and documents.

7. Electronics Manufacturing

- ❖ Promoting electronics manufacturing in India.
- ❖ Initiatives like Make in India to boost domestic production.

8. IT for Jobs

- ❖ Training and skill development for the youth in IT and electronics.
- ❖ Promoting digital literacy and IT-enabled job opportunities.

9. Early Harvest Programmes

- ❖ Implementation of short-term projects that provide immediate benefits.
- ❖ Examples include biometric attendance in government offices and Wi-Fi in universities.

✓ Major Initiatives under Digital India

- Unified Mobile Application for New-age Governance [UMANG]
- e-district mission mode project [Birth, Caste, Death, Land Records, Old Age & Widow Pension, Land Records etc.]
- Unified Payment Interface
- Digi locker
- CO-WIN
- MyGov
- DBT

- BharatNet
- Aadhar Database

➤ **Digital Initiatives for the Welfare of Farmers in India**

✓ **e-NAM (National Agriculture Market)**

- ❖ To create a unified national market for agricultural commodities by integrating existing APMC (Agricultural Produce Market Committee) mandis.
- ❖ Example - Farmers in Madhya Pradesh have reported better prices and faster payments by using the e-NAM platform to sell their produce directly to buyers across the country.

✓ **PM-KISAN (Pradhan Mantri Kisan Samman Nidhi)**

- ❖ To provide income support to all landholding farmer families to supplement their financial needs for procuring various inputs related to agriculture and allied activities.
- ❖ Millions of farmers across India have benefited from timely financial assistance, helping them purchase seeds, fertilizers, and other inputs.

✓ **Kisan Suvidha App**

- ❖ To provide information to farmers on weather, market prices, plant protection, input dealers, and agro advisories.
- ❖ Farmers in Maharashtra use the Kisan Suvidha App to check daily market prices and plan their sales, accordingly, ensuring better profits.

✓ **Pradhan Mantri Fasal Bima Yojana (PMFBY)**

- ❖ To provide a comprehensive insurance cover against failure of the crop, thus helping stabilize the income of the farmers.
- ❖ Farmers in Gujarat have benefited from prompt compensation for crop losses due to floods, enabling them to recover and prepare for the next planting season.

✓ **Agri Market App**

- ❖ To help farmers get information about the prices of crops in markets within 50 km radius of their location.
- ❖ Farmers in Rajasthan use the Agri Market App to decide the best place to sell their crops, ensuring they get competitive prices.

➤ **Digital Initiatives for Women Empowerment in India**

✓ **Mahila E-Haat**

- ❖ To provide an online marketing platform to women entrepreneurs, self-help groups (SHGs), and NGOs to showcase their products and services.

- ❖ Artisans and craftswomen from rural areas have successfully sold their handmade products through Mahila E-Haat, earning better income and gaining wider recognition.
- ✓ **Digital Saksharta Abhiyan (DISHA) or PMGDISHA**
 - ❖ To provide digital literacy to citizens, particularly focusing on women in rural areas, to enable them to use digital devices and services.
 - ❖ Women in remote villages of Bihar have learned to use digital banking and online educational resources, improving their financial management and continuing education.
- ✓ **e-SHRAM Portal**
 - ❖ To create a national database of unorganized workers, including women, to provide them with social security and welfare benefits.
 - ❖ Women domestic workers in cities like Mumbai and Delhi have registered on the e-SHRAM portal, ensuring they receive benefits like health insurance and pension.
- ✓ **Saksham Anganwadi and POSHAN Abhiyaan**
 - ❖ To use technology for monitoring and improving the nutritional status of women and children.
 - ❖ Anganwadi workers in states like Uttar Pradesh use mobile apps to track the health and nutrition of women and children, leading to better health outcomes.
- ✓ **STEP (Support to Training and Employment Programme for Women)**
 - ❖ To provide skills that give employability to women and competencies that enable them to become self-employed/entrepreneurs.
 - ❖ Women in Haryana have received digital skills training under STEP, enabling them to secure jobs in IT and service sectors.
- **Advantages of e-governance**
 - ✓ Greater Participation [RTI, MyGov.in etc.]
 - ✓ Transparency & Accountability
 - ✓ Reduces corruption & diversions
 - ✓ Enhanced accessibility [Health, education, agriculture – Blockchain, Artificial Intelligence, Machine Learning, Cloud computing etc.]
 - ✓ Increased efficiency [reduced cost, time etc.]
 - ✓ use value of information [Citizen engagement; Data Driven Approaches; reliable information within govt, between govt., citizen & business etc. can be used to address localized problems]
 - ✓ Increased economic growth [World Bank Report]
 - ✓ Environmental Impacts [Reduced Carbon Footprint]

➤ **Challenges for e-governance**

- ✓ Digital Divide & Illiteracy
- ✓ Infrastructure Limitations – Network & connectivity issues
- ✓ Cybersecurity and Privacy Concerns
- ✓ Cultural and Organizational Resistance
- ✓ Funding and Resource Constraints
- ✓ Legal and Regulatory Challenges
- ✓ Sustainability and Scalability
- ✓ User Experience and Accessibility

Questions

1. E-governance holds great promise for transforming and improving public service delivery. In this context discuss the challenges facing the digital mode of service delivery. [10]
2. Highlight some of the digital initiatives for the promotion of any two sectors in India along with the assessment of advantages of those initiatives. [10]
3. The integration of AI, Blockchain, and Data Analytics in digital service delivery to public has significantly improved efficiency, transparency, and citizen satisfaction. Illustrate with suitable examples [10]
4. Digital illiteracy and lack of ICT accessibility in rural areas significantly hinder socio-economic development. Comment [10]
5. What do you understand by e-governance? By Leveraging various models of e-governance the governments can achieve greater transparency, efficiency, and participation, ultimately promoting good governance and improving public service delivery. Discuss [15]